Question Paper ID: 545611

Course Code: IS355TBD

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

USN | 1 | R | V | 2 | 3 | C | S | 1 | 0 | 0 |

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

COMMON TO CD / CS / IS

Cloud Computing

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Explain the role of Open Standards in multi-cloud environments.	02	1	2
1.2	Differentiate between scalability and elasticity.	02	1	2
1.3	A company needs to monitor all user actions on sensitive files to comply with regulatory requirements. Which logging type applies?	02	4	3
1.4	State any two benefits of virtualization.	02	2	2
1.5	What is the role of Bungee Connect in cloud development?	02	3	2
1.6	Mention two services provided by Google App Engine.	02	1	1
1.7	Identify the use of machine images in virtualization.	02	3	2
1.8	A business stores sensitive client data in the cloud and uses encryption and access control to protect it. Which principle is being followed?	02	3	3
1.9	Name any two examples of network services.	02	3	1
1.10	What is the relationship between SOA and cloud computing?	02	2	1

Part B

Question No	Question	M	CO	BT
2a	A company that originally planned its web based system to support 10000 users suddenly notices that there is a four times increase in demand. Assuming that the company has deployed its system in a true cloud environment, discuss what are the incremental maintenance costs to adding new resources to this environment?	10	4	4
2b	Analyse how does IDaaS improve enterprise security and simplify identity management?	06	2	4
3a	Examine the role of virtual appliances within the cloud computing stack, analysing how they function as an intermediary layer between Infrastructure as a Service (IaaS) and Software as a Service (SaaS). Compare and contrast their capabilities with communication protocols and Platform as a Service (PaaS) offerings, and assess their contribution to service integration, deployment efficiency, and system manageability	10	3	4
3b	Critically analyse how the architectural concepts of pods, aggregation, and failover in an Infrastructure as a Service (IaaS) environment collectively enhance system reliability and performance. Use a labeled diagram to illustrate their interrelationships and evaluate their impact on fault tolerance, scalability, and service continuity.	06	4	3
	OR			
4a	What is IBM's Service-Oriented Modelling and Architecture (SOMA), and how does it relate to SOMF in terms of modelling based on the elements of the service life cycle?	08	3	1
4b	Interpret the various methodologies and tools used to model, manage, and optimize complex SOA-based business processes?	08	2	2
5a	A company wants to develop a real-time messaging and presence application for employees to communicate instantly and securely. i)Which protocol is suitable for this purpose? ii)Explain the main features, benefits, and drawbacks of this protocol.	08	4	3
5b	Discuss how Internet Explorer, Firefox, and Google Chrome differ in performance, user interface, and security.	08	2	2
	OR			
6a	A video conferencing application is deployed on the cloud and must support thousands of concurrent users with	08	3	2

	minimal delay. Explain how cloud networking and cloud services support high performance and scalability.			
6b	Compare and contrast Basic Public Internet, VPN, Optimized Overlay, and Accelerated Internet with respect to functionality, performance, and applications.	08	3	3
7a	A financial organization must deploy mission-critical applications requiring high security and performance. Assess the suitability of Type-1 and Type-2 hypervisors for this architecture.	10	4	5
7b	Describe hypervisors and differentiate between Type-1 and Type-2 hypervisors.	06	3	3
	OR			
8a	A cloud service provider must deploy hundreds of identical virtual machines within minutes for a new client. Evaluate the role of machine imaging in this scenario.	10	4	5
8b	Justify the use of machine imaging over manual configuration in large-scale virtual environments.	06	3	5
9a	Explain Cast Iron Cloud and its role in enterprise cloud integration.	10	4	4
9b	Explain the cloud application development lifecycle.	06	3	2
	OR			
10a	An e-commerce application hosted on the cloud experiences a major outage during a sales event. Questions: i) Identify reasons for failure. ii) Explain troubleshooting and recovery mechanisms. iii) Propose application management best practices.	10	4	5
10b	Explain the role of monitoring tools in application management.	06	2	4